

Ext and Tor

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Contents

1	Ext	2
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Let R be an (associative, unital) ring. For a cochain complex $\mathcal{C} = (C^n, d^n)_{n \geq 0}$ of R -modules, the idea of the Ext and Tor functors is to relate the cohomology of $\mathcal{C}$ to that of $\text{Hom}_R(\mathcal{C}, M)$, $\text{Hom}_R(M, \mathcal{C})$, and $\mathcal{C} \otimes M$ for some given R -module M .

1 Ext

Definition 1.1 (Projective and free resolutions) — Let A be an R -module. A *projective resolution* of A is an exact sequence

$$\cdots \longrightarrow P_3 \xrightarrow{d_3} P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} A \longrightarrow 0.$$

in which the P_n are all projective. Similarly, a *free* resolution is such an exact sequence where the P_n are all free.

Every R -module has a free (hence projective) resolution: let P_0 be the free module on A (and d_0 the obvious projection), and for $n \geq 1$ let P_n be the free module on $\ker d_{n-1}$ with d_n being the obvious projection onto $\ker d_{n-1}$. This is often unwieldy; if one has more information about A , there are often nicer resolutions one can construct.

Example 1.2. If A is projective, then it has a projective resolution

$$0 \longrightarrow A \longrightarrow A \longrightarrow 0.$$

Example 1.3. Suppose R is a PID and $A = R^n \oplus \bigoplus_{i=1}^m R/(p_i^{e_i})$ is a finitely generated R -module. There is a projection $p: R^{m+n} \rightarrow A$ with kernel $\bigoplus_{i=1}^m p_i^{e_i} R \cong R^m$, so that A has a projective resolution

$$0 \longrightarrow R^m \longrightarrow R^{m+n} \xrightarrow{p} A \longrightarrow 0.$$

For a given projective resolution P_n of A , apply the contravariant Hom functor $\text{Hom}_R(-, B)$ with a fixed R -module B to get the cochain complex

$$0 \longrightarrow \text{Hom}_R(A, B) \xrightarrow{d_0} \text{Hom}_R(P_0, B) \xrightarrow{d_1} \text{Hom}_R(P_1, B) \xrightarrow{d_2} \cdots,$$

where the morphisms are the ones that the functor induces from the morphisms of the resolution.

Definition 1.4 (Ext) — The module $\text{Ext}_R^n(A, B)$ is the cohomology of the truncated cochain complex

$$0 \longrightarrow \text{Hom}_R(P_0, B) \xrightarrow{d_1} \text{Hom}_R(P_1, B) \xrightarrow{d_2} \cdots,$$

at $\text{Hom}_R(P_n, B)$.

Example 1.5. Since $\text{Hom}_R(-, B)$ is left-exact, $\text{Ext}_R^0(A, B) = \ker d_1 = \text{im } d_0 \cong \text{Hom}_R(A, B)$.

Example 1.6. For the free resolution of $\mathbb{Z}/n\mathbb{Z}$

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

and an arbitrary abelian group B , the corresponding cochain complex is

$$0 \longrightarrow \underbrace{\text{Hom}(\mathbb{Z}, B)}_{\cong B} \xrightarrow{\cdot n} \underbrace{\text{Hom}(\mathbb{Z}, B)}_{\cong B} \longrightarrow 0,$$

so the Ext groups are

$$\text{Ext}_{\mathbb{Z}}^n(\mathbb{Z}/n\mathbb{Z}, B) = \begin{cases} {}_n B & n = 0 \\ B/nB & n = 1 \\ 0 & n \geq 2 \end{cases}$$

where ${}_n B$ is the set of n -torsion elements of B (since each element of the Hom set is uniquely determined by the image of $1 \in \mathbb{Z}/n\mathbb{Z}$, which can map to any n -torsion element).

pf

Of course, the definition of $\text{Ext}_R^n(A, B)$ a priori requires a choice of resolution, but it turns out (as the notation suggests) that it is independent of the resolution.

Besides computing projective resolutions, the other main computational tool for Ext modules is the long exact sequence in cohomology:

Theorem 1.7 (Ext long exact sequence). Let $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$ be a short exact sequence of R -modules. There is an exact sequence

$$0 \rightarrow \text{Hom}_R(N, B) \rightarrow \text{Hom}_R(M, B) \rightarrow \text{Hom}_R(L, B) \\ \xrightarrow{\delta_0} \text{Ext}_R^1(N, B) \rightarrow \text{Ext}_R^1(M, B) \rightarrow \text{Ext}_R^1(L, B) \xrightarrow{\delta_1} \dots$$

Proposition 1.8. For any R -module Q the following are equivalent:

- (1) Q is injective;
- (2) $\text{Ext}_R^1(A, Q) = 0$ for all R -modules A ; and
- (3) $\text{Ext}_R^1(A, Q) = 0$ for all R -modules A and all $n \geq 1$.

A similar construction can be done with the covariant Hom functor, but we need a different kind of resolution, since projective resolutions don't allow us to take advantage of left-exactness here. Therefore, the obvious choice is to dualize and take an *injective resolution*; as the name suggests, this is an exact sequence

$$0 \longrightarrow A \xrightarrow{d_0} Q_0 \xrightarrow{d_1} Q_1 \xrightarrow{d_2} Q_2 \xrightarrow{d_3} Q_3 \longrightarrow \dots$$

in which the Q_n are all injective R -modules.